



Green bricks

As strong as concrete, a soil composite-based building block by Integrity takes 40 per cent less energy to make and contains 50 per cent pre-recycled materials



Solar balloons

Imagine shiny balloons that go up into the sky and generate gigawatts of power. Find out more at www.solar-balloons.com

Tomorrow's tech

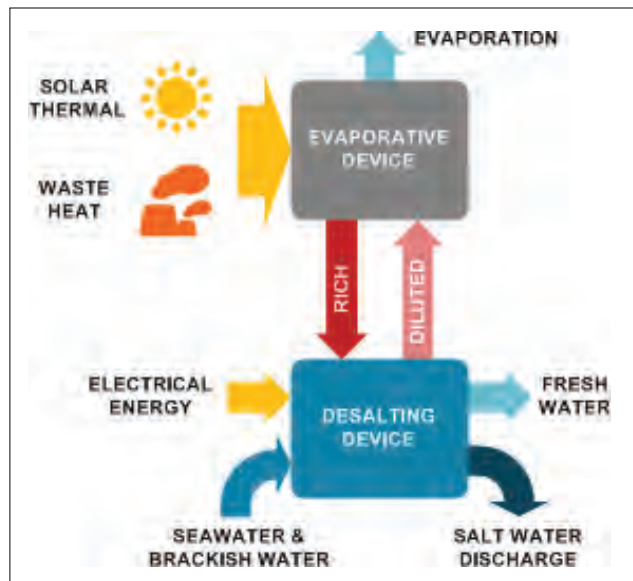
fuel cell that utilises a stack of ceramic plates, each coated with two different proprietary chemical mixes which Sridhar calls 'inks' as the electrolyte. Unlike other fuel cells that need to be fed pure hydrogen, the BloomBox uses natural gas. It can even use green fuels like bio gas and landfill gas. It is more efficient at converting the fuel into energy – using only half the amount of gas as traditional power plants would use to produce the same amount of electricity and uses low cost materials even for the catalyst, which in most cases is platinum.

These BloomBoxes can be linked together to combine their energy output. His bigger boxes or Energy Servers, which are about the size of a parking space have been secretly powering some of the most well known companies in Silicon Valley for quite some time. eBay is reportedly using the 100 Kilowatt boxes for powering 15 per cent of its headquarters.

Currently, even with the subsidies and taxbreaks, the costs are still not favourable for large scale implementation. They have a significant investment and the ceramic plates need replacing every five years. Still the company believes the technology will be brought down to within the reach of an average home owner over time. Sridhar's powerplant in a box could one day take you off the electricity grid entirely, since his dream is to install one in every backyard. "Don't start signing up for orders yet, this is a product of the future", he said at the unveiling.

Solar-powered water de-salination

According to Karan, head-business development at Shashwat Green Fuels & Technologies "Our biggest challenge is Water and not Energy. People can live without lights or air conditioners or an iPod, but they cannot survive without water." Although most of Mankind is located near water, not much of it is usable for drinking. 97 per cent of all water on earth is sea water and half of what is available is frozen at the poles. By some estimates nearly 20 per cent



The desalination process explained

of humanity does not have access to clean drinking water. The quest for desalinating or removing salt from sea water to make it drinkable has been on for quite some time. Some of the well-known methods of desalinating water are distillation or reverse osmosis however, as you can imagine they are very energy intensive and hence expensive. So what about the majority of the world's population that cannot afford this luxury?

A company called "Saltworks Technologies" from Vancouver, Canada, may have the solution. The company is on the verge of commercialising a breakthrough desalination technology during a time of increasing freshwater scarcity, rising energy prices, and mounting concerns over carbon impacts. The company has come up with Thermo-ionic technology that can make cheap drinking water from the sea using evaporation and ionic conduction, powered mainly by sunlight. Saltwater is evaporated to produce a concentrated solution.

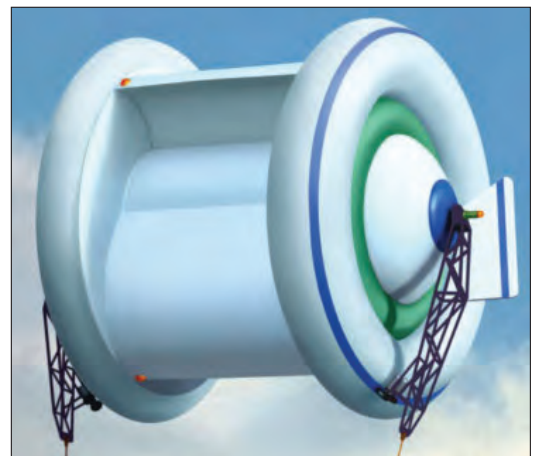
This solution, which has concentration gradient energy, is fed into Saltworks' proprietary desalting device to desalinate either seawater or brackish water. Some electrical energy is used to circulate fluids at a low pressure, yet the bulk of the energy input is obtained through the evaporation itself. The

amazing features of this kind of an approach are that it uses up to 80 per cent less electrical/mechanical energy that is required by plants based on reverse-osmosis and other processes. Theoretically, the technology is scaleable enough to supply anything from cities to small villages.

High altitude wind power generation

Wind, along with the sun, have always been looked at as the prime sources of renewable energy. There has been a lot of work in the solar sector in both photovoltaics (PV) and concentrating solar power (CSP). A company called

Nanosolar has even come up with technology to make super thin photovoltaic films. Still, one of the most promising sources and relatively untapped is Wind. Magenn Power is making strides in this sphere. "There is enough energy in high altitude winds to power civilisation 100 times over; and sooner or later, we're going to learn to tap into the power of winds and use it to run civilization." says Ken Caldeira, Professor of Global Ecology at the Carnegie Institution for Science on Magenn Power's web site. Magenn Power has developed a high altitude wind turbine system called MARS - short for Magenn Air Rotor System. It's essentially a flying turbine in the form of a large balloon which spins at high altitudes generating power. "MARS is a lighter-than-air tethered wind turbine that rotates about



The Magenn high altitude turbine prototype